A PATH-GRAPH-BASED COMBINATORIAL IDENTITY

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August 9, 2025

ABSTRACT

Using path graphs, I show the combinatorial equality:

$$\sum_{i=0}^{n-1} \binom{n-1}{i} (n-i) = 2^{n-1} + \sum_{j=1}^{n-1} \sum_{i=0}^{j-1} \binom{j-1}{i} (j-i)$$

Keywords Combinatorial identities, Path graphs.

1 Introduction

In a previous work [1], I focused on Bernoulli graphs to obtain several combinatorial identities. In here, I use path graphs. The general approach remains similar: describe a graph by two different means, and then match the two analytical expressions. In the previously cited work, I described a particle whose dynamics followed probabilistic laws. The probabilities of having attained a barrier were equal to the probability of having arrived earlier plus the probability of still being elsewhere than in the barrier. In this work, the recursive structure of path graphs of length n settles the equality. Separately, we infer easily the analytical form.

2 Path graphs of length n

Definition 2.1. *Path graphs:*

A path graph of length n is is a graph whose vertices can be listed in the order v_1, v_2, ..., v_n such that the edges are $\{v_i, v_{i+1}\}$ where $i = 1, 2, ..., n-1$. [1]. *In here we consider path graph whose*

T_n is a collection of branches, every branch has length n, $n \in \mathbb{N}$. Every branch of the tree describes one way of attaining n by summing integer numbers.

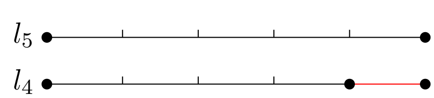
Definition 2.2. *Concatenation of a segment and a tree:*

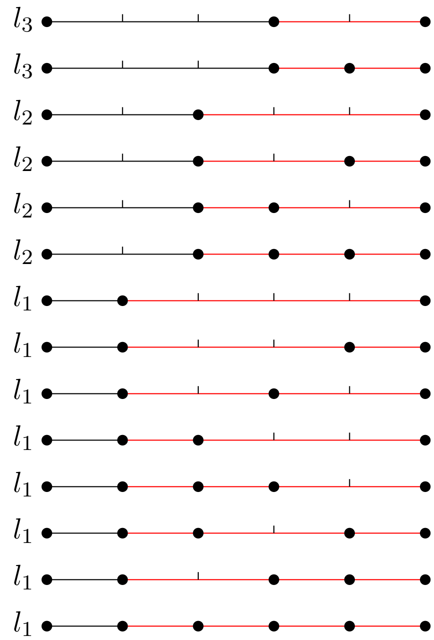
We can concatenate a segment of length j to a tree T_n. It results in a tree with n branches, and each branch is composed of a segment of length j and one of the n branches of the tree T_n.

Proposition 2.3. *Recursive definition of a tree: A tree of length n can be defined recursively as:*

$$T_n = \sum_{i=0}^{n-1} l_{n-i}.T_i$$

Proof. Case n = 5.





□

Definition 2.4. *Number of branches:*

$c(T_n)$ is the number of branches of a tree of size n. For instance $c(T_1) = 1, c(T_2) = 2, c(T_3) = 4$. By convention we take $c(T_0) = 1$.

Proposition 2.5. *Number of branches (recursive definition):*

$$\begin{cases} c(T_0) = 1 \\ c(T_n) = \sum_{i=0}^{n-1} c(T_i), \quad \textit{for } n \geq 1 \end{cases}$$

Proof. Using the recursive definition of a tree:

$c(T_n) = \sum_{i=0}^{n-1} c(l_{n-i}T_i) = c(l_nT_0) + \sum_{i=1}^{n-1} c(l_{n-i}T_i) = 1 + \sum_{i=1}^{n-1} c(T_i) = \sum_{i=0}^{n-1} c(T_i)$

□

Proposition 2.6. *Number of branches (direct value):*

$$\begin{cases} c(T_0) = 1, & \textit{for } n = 0 \\ c(T_n) = 2^{n-1}, & \textit{for } n \geq 1 \end{cases}$$

Proof. From the previous graph (n=5) we see that for each position (excepting the initial and the final) there are two possibilities. Therefore $c(T_n) = 2^{n-1}$.

Using the recursive definition of this graphs, we see that this approach is correct:

$c(T_n) = \sum_{i=0}^{n-1} c(l_{n-i}T_i) = c(l_nT_0) + \sum_{i=1}^{n-1} c(l_{n-i}T_i) = 1 + \sum_{i=1}^{n-1} c(T_i) = 1 + \sum_{i=1}^{n-1} 2^{i-1} = 1 + \sum_{i=0}^{n-2} 2^i = 2^{n-1}$.

□

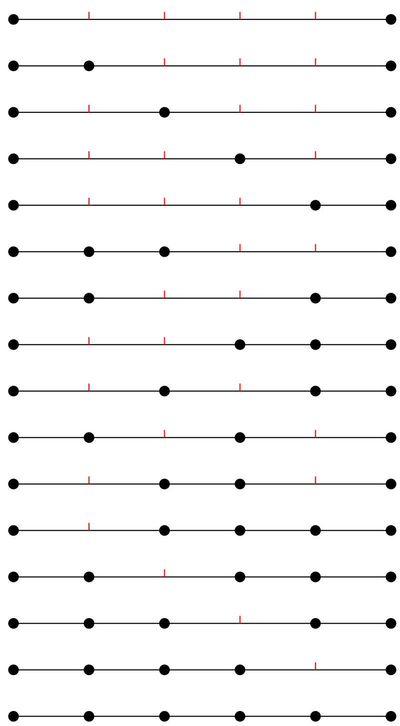
Definition 2.7. *Let l_i a graph composed of two vertices, the initial and the final. We define $s(l_i) = 1$. Starting from the initial vertex and stopping at the final vertex, there is one stop. Let $N(T_n)$ be the total number of vertices in a tree T_n. In a tree T_n, composed of $c(T_n)$ branches, there is $s(T_n)$ stops. $s(T_n) = N(T_n) - c(T_n)$.*

Proposition 2.8. *Number of stops:*

$$s(T_n) = \sum_{i=0}^{n-1} \binom{n-1}{i} (n-i)$$

Proof. We can redraw a tree of order n grouping the branches by the number of empty spots.

Case n= 5:



Therefore the total number of stops is equal to the sum of the number of branches with i empty spots times the number of stops each of these branches of i empty spots.

□

Proposition 2.9. *Number of stops (recursive relationship):*

$$s(T_n) = 2^{n-1} + \sum_{i=1}^{n-1} s(T_i)$$

Proof.

$$\begin{aligned} s(T_n) &= s(\sum_{i=0}^{n-1} T_i.l_{n-i}) \\ &= 1 + s(\sum_{i=1}^{n-1} T_i.l_{n-i}) \\ &= 1 + \sum_{i=1}^{n-1} c(T_i) + \sum_{i=1}^{n-1} s(T_i) \\ &= 1 + (2^{n-1} - 1) + \sum_{i=1}^{n-1} s(T_i) \end{aligned}$$

□

Lemma 2.10.

$$\sum_{i=0}^{n-1} \binom{n-1}{i} (n-i) = 2^{n-1} + \sum_{j=1}^{n-1} \sum_{i=0}^{j-1} \binom{j-1}{i} (j-i) \tag{1}$$

Proof. Substitute the analytical expression $s(T_n)$ into it's recursive form. □

3 Conclusion

There are several extensions for this approach, in particular for latices of dimension n. Current efforts aim to describe the combinatorics of two-dimensional lattices.